

Software Systems Design

(Architecture · 2024)

Outline

— [Software Architecture

— [Quality Attributes

— [Architecture Patterns

— [Designing Software
Architecture

— [Documenting Software
Architecture

— [Evaluating Software
Architecture

— [Microservices

Software Architecture in General

- [What is software architecture?

- Structure, Elements, Relationships, Design

- [What does a software architect do? Liaison, Software engineering, Tech. knowledge, Risk mgmt.

- [Where do architectures come from?

- NFRs, ASRs, Quality Requirements; Stakeholders, Organisations, Technical Environments...

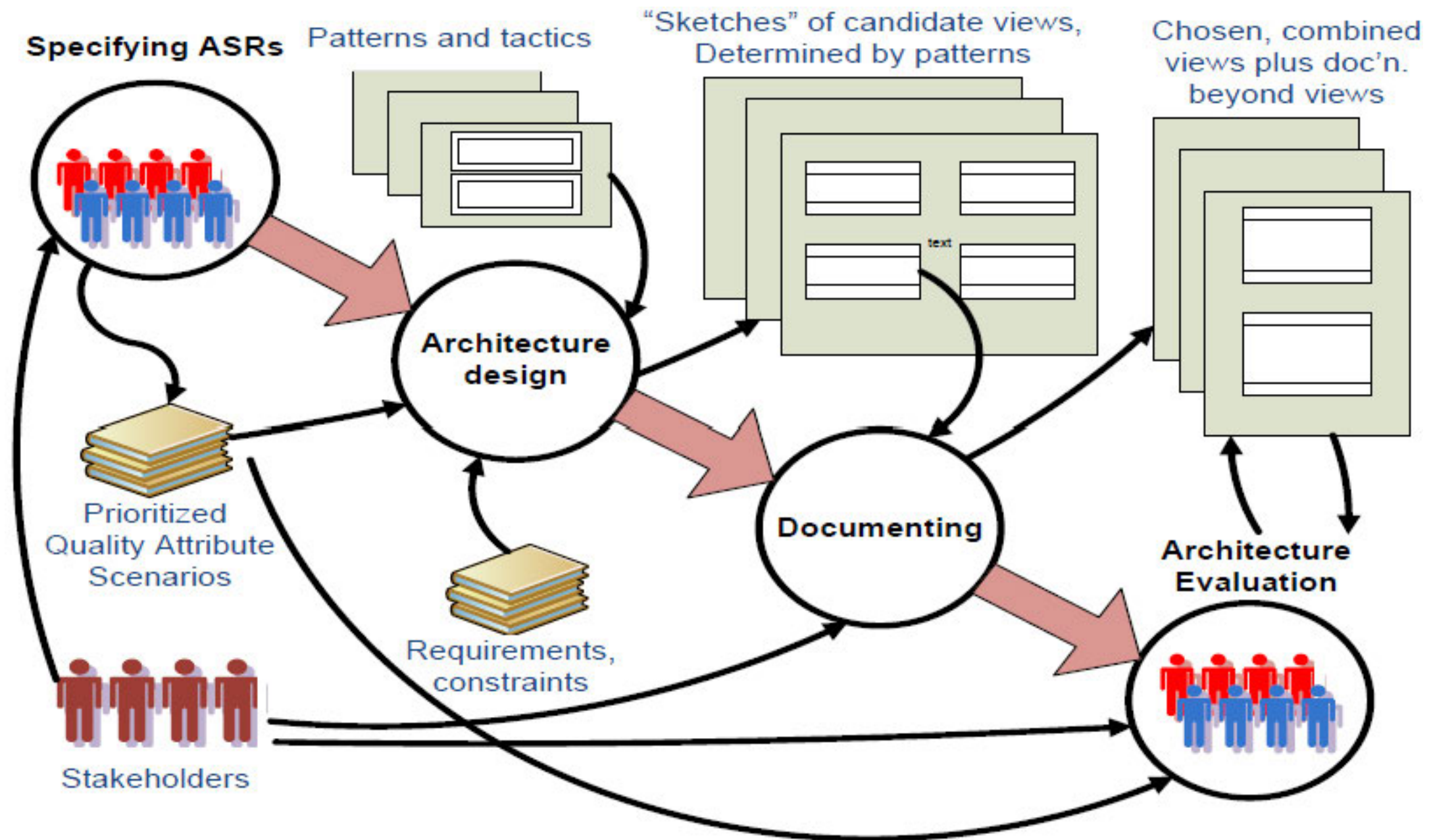
- [Architecture (4+1) Views

- Logical view, Process view, Physical view, Development view + Use case scenarios...

- [Architectural activities and process

- [Software architecture knowledge areas

Architecture Process



Quality Attributes

Software Requirements

- Functional requirements, Quality requirements (NFRs), Constraints

Quality Attributes

- Internal vs. External Attributes
- Modeling quality attribute scenarios: Source, Stimulus, Artefact, Environment, Response, Measure
- Availability, Interoperability, Modifiability, Performance, Security, Testability, Usability, X-ability...
- Tactics for quality attributes
- Checklist for architecture design decisions

Architecturally Significant Requirements

- How to gather and identify ASRs: Requirements, Workshop (interviews), Business goals, Utility tree

Architecture Patterns

Architecture Patterns

- Context, Problem, Solution: elements + relations + constraints

Module Patterns

- Layered pattern (micro-kernel pattern)

Component-Connector Patterns

- Broker pattern, Model-view-controller pattern, Pipe-and-filter pattern, Client-server pattern, Peer-to-peer pattern, Service-oriented pattern, Publish-subscribe pattern, Share-data pattern

Allocation Patterns

- Map-reduce pattern, Multi-tier pattern

Patterns vs. Tactics

Designing Architecture

General Design Strategy

- Abstraction, Decomposition, Divide & conquer, Generation and test, Iteration, Reuse

Categories of design decisions

- Responsibilities, Coordination, Data, Resources, Elements mapping, Binding time, Technology

Attribute-Driven Design (ADD 2.0)

- Inputs to and outputs of ADD
- 8-step process: 1. confirm requirements, 2. choose an element to decompose, 3. identify ASRs, 4. choose a design satisfying ASRs, 5. instantiate elements & allocate responsibilities, 6. define interface, 7. verify & refine requirements, 8. repeat step 2-7 until all ASRs satisfied
- Step4: 4.1 identify concerns, 4.2 list alternatives, 4.3 select patterns/tactics, 4.4 determine relations, 4.5 capture views, 4.6 resolve inconsistencies

Documenting Architecture

Views and Beyond

Views:

- Styles (viewpoints), patterns and views

- Structural views: module views, component-and-connector views, allocation views

- Quality views

- Documenting views: 1. build stakeholder/view table, 2. combine views, 3. prioritise & stage

- Info. beyond views: documentation info & architecture info (mapping between views)

- Documentation package: views + beyond

Evaluating Architecture

- [Architecture analysis & evaluation methods

- [ATAM: Architecture Tradeoff Analysis Method

- Stakeholders involved in ATAM

- Inputs to and Outputs of ATAM: Risks (Non-risks, Risk themes), Sensitivity points, Trade-off points

- Phase 0: Partnership & preparation; Phase 3: Follow-up

- Phase 1: Evaluation - 1

- 1. present ATAM, 2. present business drivers, 3. present architecture, 4. identify architectural approaches, 5. generate utility tree, 6. analyse architectural approaches

- Phase 2: Evaluation - 2

- 1. present ATAM & results, 7. brainstorm & prioritize, 8. analyse architectural approaches, 9. present results

Final Exam

——[总评成绩 = 平时成绩 * 40% + 期末考试 * 60%

——[内容占比： 架构设计50% + 设计模式50%

——[题型： 简答题、设计分析题

——[英文题目、中文或英文答题

——[个别题目可能需画图

——[基础内容60%

——[高阶内容40%



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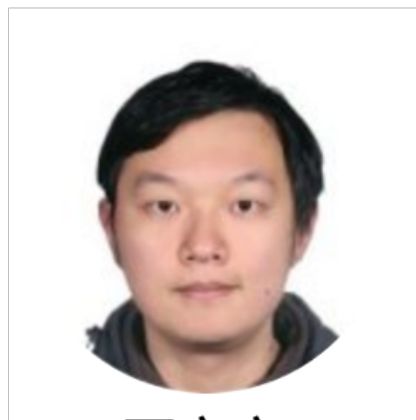
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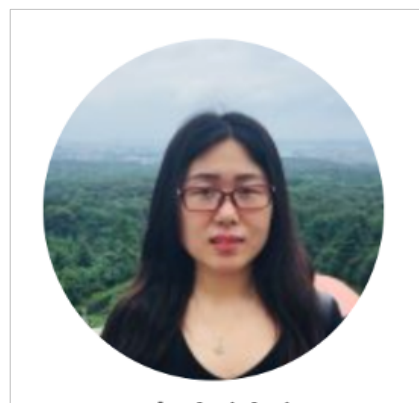
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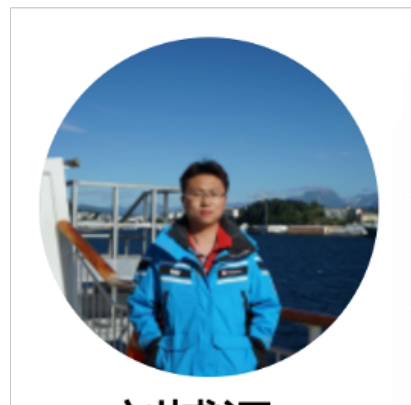
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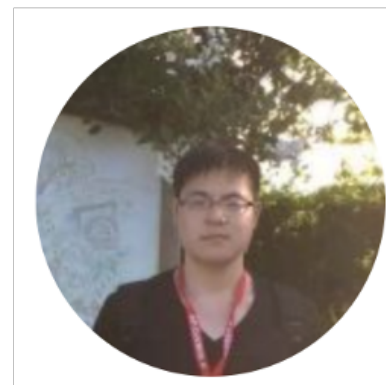
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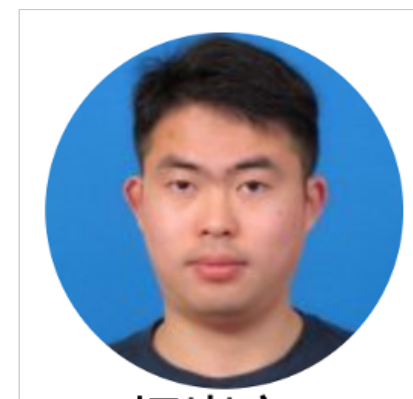
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